

AIC, TIC, and RMSEA as Bias Corrections

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Purpose

This note records the common shape behind AIC/TIC and RMSEA. Both start from a sample criterion that is too favorable because it is evaluated after fitting. Both subtract a first-order bias term. The bias terms are different because the targets are different:

AIC/TIC: predictive risk of the fitted distribution,

RMSEA: population model misfit, or noncentrality per observation per df.

AIC and TIC do not require the exact-fit or local-alternative story that often motivates RMSEA. They use local Taylor expansions around the fitted parameter and its pseudo-true target, but the statistical target is fixed-misspecification predictive risk, not a Pitman drift from a correct model.

1 AIC: correcting in-sample optimism

Let $Y_1, \dots, Y_n \sim g$ and let f_θ be a parametric working model. Write the log likelihood

$$\ell_n(\theta) = \sum_{i=1}^n \log f_\theta(Y_i), \quad \hat{\theta} = \arg \max_{\theta} \ell_n(\theta).$$

The population target of maximum likelihood, even when the model is wrong, is the KL projection

$$\theta_0 = \arg \max_{\theta} M(\theta), \quad M(\theta) = \mathbb{E}_g \{\log f_\theta(Y)\}.$$

Equivalently, θ_0 minimizes $\text{KL}(g, f_\theta)$ over the model class.

The quantity we would like to estimate is the expected log score of the fitted model on a fresh observation,

$$R_n = \mathbb{E}_{\hat{\theta}} \mathbb{E}_{Y^0} \{\log f_{\hat{\theta}}(Y^0)\},$$

where $Y^0 \sim g$ is independent of the training sample and the outer expectation averages over the training sample. The empirical maximized log likelihood $n^{-1}\ell_n(\hat{\theta})$ estimates the same log score on the training observations, not on fresh observations. It is optimistic because $\hat{\theta}$ was chosen to fit those same observations.

Proposition 1 (Akaike's optimism correction). *If the model is correctly specified, regular, and has k free parameters, then*

$$\mathbb{E}\{\ell_n(\hat{\theta})\} = n \mathbb{E}\{\log f_{\hat{\theta}}(Y^0)\} + k + o(1).$$

Therefore

$$\ell_n(\hat{\theta}) - k$$

is an asymptotically unbiased estimator, up to constants independent of the candidate model, of the expected out-of-sample log likelihood. Multiplying by -2 gives

$$\text{AIC} = -2\ell_n(\hat{\theta}) + 2k.$$

Derivation sketch. Let

$$s_i(\theta) = \frac{\partial \log f_{\theta}(Y_i)}{\partial \theta}, \quad I = -\mathbb{E} \left\{ \frac{\partial^2 \log f_{\theta_0}(Y)}{\partial \theta \partial \theta^\top} \right\}.$$

Under correct specification, $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I^{-1})$ and $\mathbb{E}\{s_i(\theta_0)s_i(\theta_0)^\top\} = I$.

For a fresh observation,

$$M(\hat{\theta}) = M(\theta_0) - \frac{1}{2}(\hat{\theta} - \theta_0)^\top I(\hat{\theta} - \theta_0) + o_p(n^{-1}),$$

so

$$n \mathbb{E}M(\hat{\theta}) = nM(\theta_0) - \frac{1}{2}k + o(1).$$

For the training log likelihood, the score term does not average away because $\hat{\theta}$ is a function of the same data. A quadratic expansion of $\ell_n(\hat{\theta})$ around θ_0 gives

$$\mathbb{E}\ell_n(\hat{\theta}) = nM(\theta_0) + \frac{1}{2}k + o(1).$$

The gap is $k + o(1)$ on the log-likelihood scale, or $2k + o(1)$ on the deviance scale. This is the AIC penalty. $\square$

Remark. AIC is often described as making the maximized log likelihood approximately unbiased for expected predictive log likelihood. More precisely, it removes the first-order *optimism* of reusing the training data for both fitting and evaluation. It is not a small-sample unbiasedness claim, and it is not a test of exact fit.

2 TIC: the same correction under misspecification

When the model is misspecified, the Hessian and score variance no longer agree. Define

$$I = -\mathbb{E}_g \left\{ \frac{\partial^2 \log f_{\theta_0}(Y)}{\partial \theta \partial \theta^\top} \right\}, \quad J = \mathbb{E}_g \{ s(Y; \theta_0) s(Y; \theta_0)^\top \}.$$

Then

$$\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, I^{-1} J I^{-1}).$$

The same local expansion gives the optimism

$$\text{tr}(I^{-1} J)$$

on the log-likelihood scale. Takeuchi's information criterion is therefore

$$\text{TIC} = -2\ell_n(\hat{\theta}) + 2 \text{tr}(\hat{I}^{-1} \hat{J}).$$

Correct specification is the special case $I = J$, giving $\text{tr}(I^{-1} J) = k$ and recovering AIC.

Remark. This is the sense in which AIC/TIC use local arguments indirectly. The target is fixed-population predictive risk, but the bias calculation is a second-order expansion around the KL projection. No local drift toward a true finite dimensional model is needed.

3 RMSEA: correcting chi-square noise

Now let $\hat{\eta}$ be a consistent estimator of a saturated population object η , such as (μ, Σ) or a limited-information ordinal moment vector. Let $\mathcal{C} \subset E$ be the model-implied class and let

$$F_{\mathcal{C}}(\eta) = \inf_{c \in \mathcal{C}} D(\eta, c)$$

be a nonnegative discrepancy, with $F_{\mathcal{C}}(\eta) = 0$ iff $\eta \in \mathcal{C}$. Continuity gives

$$F_{\mathcal{C}}(\hat{\eta}) \xrightarrow{p} F_{\mathcal{C}}(\eta)$$

even when $\hat{\eta}$ has finite-sample bias. The population RMSEA target is

$$\varepsilon_{\mathcal{C}}(\eta) = \sqrt{\frac{F_{\mathcal{C}}(\eta)}{df}}.$$

The usual SEM test statistic has the form

$$T_n = nF_{\mathcal{C}}(\hat{\eta})$$

after fitting the projection onto $\mathcal{C}$; in magmaan's convention $T_n = 2n \mathbf{fmin}$. Under exact fit and the usual regularity conditions,

$$T_n \xrightarrow{d} \chi_{df}^2,$$

so pure sampling noise contributes about df to the statistic even when the population discrepancy is zero. Under fixed misspecification,

$$T_n = nF_{\mathcal{C}}(\eta) + O_p(\sqrt{n}),$$

so the exact-fit test diverges, but the rescaled discrepancy remains meaningful. The RMSEA plug-in correction is

$$\hat{\varepsilon} = \sqrt{\frac{\max(T_n - df, 0)}{n df}}.$$

Remark. The subtraction of df in RMSEA is analogous in spirit to the AIC subtraction of k : both remove a first-order fitting/noise contribution. But the targets are not the same. AIC/TIC correct an in-sample log score to estimate out-of-sample predictive risk. RMSEA corrects a chi-square-like lack-of-fit statistic to estimate excess noncentrality, hence population discrepancy per degree of freedom.

4 Where local alternatives enter

The population definition

$$\varepsilon_{\mathcal{C}}(\eta) = \sqrt{F_{\mathcal{C}}(\eta)/df}$$

does not require local alternatives. It is a fixed-population estimand. Local alternatives enter when one wants the chi-square reference distribution,

confidence intervals, close-fit tests, or power calculations to remain nondegenerate. A Pitman sequence writes

$$\eta_n = \eta_0 + h/\sqrt{n}, \quad \eta_0 \in \mathcal{C},$$

so

$$nF_{\mathcal{C}}(\eta_n) \rightarrow \lambda \quad \text{and} \quad T_n \xrightarrow{d} \chi_{df}^2(\lambda).$$

Then RMSEA is the square root of $\lambda/(n df)$ on the sample scale, or of $F_{\mathcal{C}}(\eta_n)/df$ on the population-discrepancy scale. Without the local sequence, fixed positive misfit makes T_n diverge, so there is no finite noncentral chi-square limit to invert; there is still a perfectly good population discrepancy.

5 Comparison

criterion	target	bias/noise correction
AIC	predictive log likelihood, correct model	k on log-likelihood scale
TIC	predictive log likelihood, misspecified model	$\text{tr}(I^{-1}J)$
RMSEA	population discrepancy per df	df chi-square noise

Thus the closest one-line memory aid is:

AIC/TIC estimate prediction risk after fitting;

RMSEA estimates lack of fit after removing expected test noise.